Secure and Efficient Fast Initial Link Setup Scheme Design

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***Abstract*—With the increasing demands for fast and secure wireless link connections to the access points (APs) supporting large quantities of devices in wireless local networks (WLANs), Fast Initial Link Setup (FILS) has been proposed. It is the initial link setup mechanism specified in IEEE 802.11ai, which is a new amendment to IEEE 802.11 standard family to support massively deployed wireless nodes. However, security concerns on the link connection have not been fully eliminated, especially for the authentication process. For example, a type of recently revealed malicious attack, Key Reinstallation Attack (KRA) might be a threat to the FILS authentication. To prevent the success of the KRAs, in this paper, we propose a Secure and Efficient FILS (SEF) protocol as its optional substitute. The SEF is designed to eradicate potential threats from the KRAs without degrading the network performance.**

Keywords—IEEE 802.11ai; Fast Initial Link Setup; Key Reintallation Attack

I. INTRODUCTION

The number of mobile devices has been drastically increasing in recent years, which has brought not only the convenience and ubiquitous access to the Internet, but also other challenges. The question is how to improve the overall performance when a large amount of devices are accessing the network within a short period. With massive wireless devices attempting to access Wireless Local Area Networks (WLANs), the network needs a new standard to limit the latency to an acceptable level. Thus, the Fast Initial Link Setup (FILS) has been specified in IEEE 802.11ai standard [1] to answer the challenge. The FILS intends to limit the link setup time to less than 100ms. According to [2], it has introduced the FILS discovery frame to allow more frequent and efficient beacons and network detection. Neighbor Access Point (AP) and the FILS parameter advertisements have been included to accelerate the AP handovers. Many new features have been added into the FILS to improve the performance of the networks, while it has also enhanced the security functionality with the new authentication and association schemes. The FILS can effectively complete the authentication and association processes as well as high layer operations such as IP address assignment with two round-trip message exchanges [1].

The FILS, as discussed in [3], has four steps of operations to complete the link set up, which are AP discovery, network discovery, authentication and association, and higher layer operations. Some possible improvements on the discovery algorithms of the FILS have been suggested in [3] without a detailed discussion on the authentication and security issues. The FLAP proposed in [4] is the primitive version of the FILS, and it was published before the FILS with some new features shown in the FILS. Thus, we decide to work based on the FILS instead of the FLAP scheme. Some potential denial of service (DoS) attacks against the FILS authentication scheme have been introduced and discussed in [5]. These attackers are able to forge the messages to fool the victims to accept and process too many messages so that the wireless connection could not be established. However, by the protection of Extensible Re-authentication Protocol (ERP) [6], who defined the information exchange scheme of the first two messages in FILS, some of these attacks are actually infeasible. The DoS attacks mentioned in [5] are indeed the threats to the security of the protocol. Nevertheless, in those published works, there are only a few aiming at addressing the security issues of the FILS authentication. Even none of the works has detailed discussions on the threats from the Key Reinstallation Attacks (KRAs) to the FILS authentication. The KRA is a new type of malicious attacks to the network protocols described in [7], which is able to jeopardize the security of various WLAN authentication protocols, such as 4-Way Handshake, Group Key Handshake, Fast Transition (FT) Basic Service Set (BSS), and Peer-Key Handshake. Since the FILS authentication is designed based on the previous authentication protocols specified in IEEE 802.11i standard, the FILS is much likely to be vulnerable to the KRAs.

In this paper, we explore the vulnerability of the FILS authentication under the KRAs. Based on the analysis, we propose the Secure and Efficient FILS (SEF) to defend the FILS authentication against KRAs. To the best of our knowledge, this is the first piece of the research work to address the KRAs to the FILS authentication in WLANs with an effective and efficient solution to protect the FILS authentication from the KRAs.

The rest of the paper is organized as follow: Section II is a review of original FILS authentication. Section III presents the details of the proposed solution. Section IV and V give the security analysis and performance evaluation of the SEF respectively. The last section presents the conclusion.

II. REVIEW OF FILS AUTHENTICATION AND KRA

A. FILS Authentication

The FILS authentication has two options for a station (STA) to choose including the shared key authentication and the public key authentication. Since the public key authentication has a higher computation complexity and our design intends to improve the resource-constrained networks, the proposal is designed based on the shared key authentication, and only the shared key authentication is reviewed in this paper. When a STA receives a beacon or a probe response from the AP, it may initiate an authentication process by sending the first message. The messages exchanged between the AP and the STA are denoted as Msg1, Msg2 and so forth. In Msg1, the STA indicates which option it chooses. If the STA believes that it shares the re-authentication Root Key (rRK) with the Authentication Server

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(AS) from a full 802.1x authentication, it will initiate the shared key authentication.

Since the proposal is designed to address protocol flaws between the STA and the AP, it is assumed that the message exchange between the AP and the AS goes through a secure channel. The first two messages exchanged between the AP and the STA and the messages exchanged between the AP and the AS are specified by the ERP in [5]. The ERP describes the way for the STA and the AS to authenticate each other and the way for the STA and the AP to derive the common secret key from the parent key rRK. The common secret key is re-authentication Master Session Key (rMSK). Thus, basically, Msg1 and Msg2 are used for the derivation of the common session keys such as Pairwise Master Key (PMK) and Pairwise Transient Key (PTK), to be used for data transmission. To derive the session keys, the STA can either attempt Pairwise Master Key Security Association (PMKSA) or calculate new PMK with rMSK and nonces. After the derivation of PMK and PTK, the association messages will be used for confirmation and IP address assignment. By calculating the Key-auth field in the FILS Key Confirmation element with hash functions, the STA and the AP are able to construct the association messages and confirm the possession of keys by comparing each other's Key-auth field, which is calculated with Integrity Check Key (ICK) on a concatenation of the STA's nonce, the AP's nonce, the Medium Access Control (MAC) address of the STA, and the AP's Basic Service Set Identity (BSSID). Upon the successful confirmation, the AP and the STA are able to use the session keys for data transmission.

B. ERP

The FILS shared key authentication adopts ERP as its authenticity information exchange protocol. According to key hierarchy, parent key rRK is used to derive rMSK and re-authentication Integrity Key (rIK). rMSK is for calculation of PMK, and rIK is used for Msg1 and Message Integrity Check (MIC). With the protection of rIK, the Msg1 containing the indicator of key name and the STA's nonce is forwarded to the AS to be verified by the AS's rRK. If the verification succeeds, the AS will send the AP rMSK and inform the AP to send back the success confirmation as well as the AP's nonce. And then both of them calculate the session keys with rMSK and nonces.

C. Vulnerability of the FILS under the KRAs

The threats from the KRAs to the FILS authentication are critical be explored to understand the vulnerability of the FILS. The KRA is a general type of malicious attack, which enforces a victim to reinstall the keys in use, while the way of its working is various for different protocols. To attack the FILS authentication, one possible attack strategy is similar to that of the attack against 4-Way Handshake. The KRAs on 4-Way Handshake have been discussed and classified into several categories in [7], while there is one that fits the FILS authentication described as follows.

In order to attack 4-Way Handshake, the KRA first can obtain a middle position between the STA and the AP for adversary. Then, a channel-based man in the middle (MitM) attack [8] is ready to launch. The adversary can first faithfully relay the message to both parties until the last message Msg4.

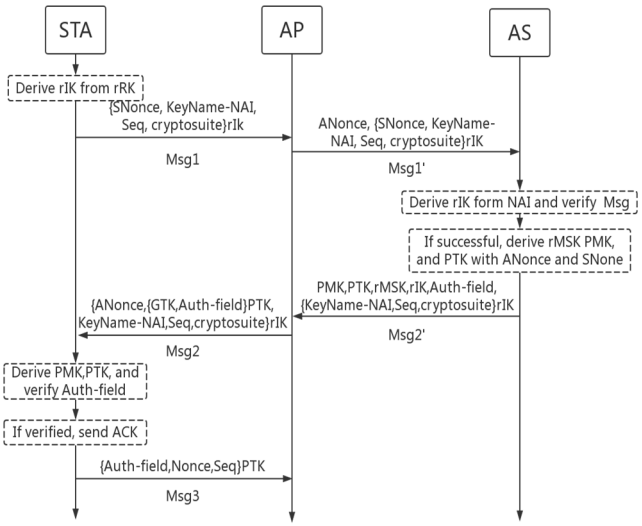


Figure 1 SEF Mechanism

Since Msg4 is intercepted by the adversary, the AP cannot receive the acknowledgement from the STA. The AP will retransmit the Msg3, which could let the STA receive the Msg3 twice and the two encrypted Msg3s will confuse the STA. Then, the STA will reinstall the key in use and reproduce a nonce. The attack has been disclosed and illustrated in [7].

III. PROPOSAL OF THE SEF

Based on the above analysis, it is clear that the reason to make the KRAs launch successfully is that the attackers can acquire a MitM position and trick the victim to install the key set twice. To avoid being confused by two consecutive messages, new algorithms to control the operations should be defined to ensure both STAs and APs strictly follow the work flow specified by the original protocol. According to the idea to block the possible success of the KRAs, we design the SEF scheme to protect the FILS shared key authentication as an alternative authentication protocol. Essentially, it should have outstanding features to be secure enough against KRAs and other typical malicious attacks. Furthermore, it should improve the efficiency of the FILS authentication by less message exchanges and reallocating the calculation from the AP to the AS. We describe the details of SEF with its message exchanges and the operations designed for the STA and the AP.

A. Message Exchange

The message exchange flow is presented as in Figure 1. The messages exchanged between the AP and the AS, denoted as Msg1, Msg2, etc. are presented below according to the message order. They are exchanged for the calculation and derivation of rMSK, PMK, and PTK by the AS for the AP. Noted that the rIK is not an encryption key so that the Msg1 and Msg1' are protected by MIC without an encryption, because the common encryption keys are not yet derived.

Msg1 and Msg1' are used in the same way as in the ERP protocol [6] where the AP as an authenticator is not able to verify rIK but the sequence number. If the monotonically increasing sequence number is larger than that before, the AP simply

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appends the ANonce and relays the Msg1 to the AS to verify the validity of rIK as well as the sequence number. The verification information, $\{GTK, Key-Auth\}_{PTK}$ is included in Msg2 for the STA to verify. This information embedded in Key-auth field is encrypted by PTK because the STA is able to derive PTK and verify the Key-auth field with rMSK and the nonce received. Upon on successful verification, Msg3 is sent as an acknowledgment the AP to report the authentication successful. The nonce in Msg3 is used to provide freshness. The protection to the replay attacks is provided by the sequence number. Note that PTK used as encryption key in the abovementioned messages implies the message is protected by authenticated encryption with associated data (AEAD) algorithm with Key Encryption Key (KEK). The SNonce is a nonce generated by the STA and the ANonce is a nonce generated by the AP. The Key-auth field is used to confirm the possession of session keys. The Seq field indicates the sequence number (initialized to zero), which is maintained corresponding to the specific keyName-NAI according to ERP [6]. They are used for the protection to replay attacks and the generation of rMSK as that used in ERP [6].

B. The Operations

In IEEE 802.11ai standard [1], since the prevention of the KRAs to the FILS authentication has not been addressed as indicated in [7], the operations of the STA and the AP leave loopholes to the KRAs. In order to prevent the KRAs with efficiency, the proposed SEF is designed to support the FILS authentication with the ability against the KRAs for the STA and the AP. We use a flowchart to describe the operations of the STA in Figure 3 in appendix, in which, the STA starts its initialization of the authentication with 5 operations. It is clear that the STA uses the information in Msg2 to confirm the possession of rRK from the AS and calculate PMK and PTK. If the STA verifies the Key-auth field successfully, it will install key set and send an acknowledgement (ACK), while if Msg2 is abused, the STA will not install the keys. Instead, it accepts the first ANonce and key set, and simply resends Msg3 as the ACK again so that the KRA could not succeed. To achieve this, the STA can use a Boolean value as a flag to indicate that the key set is installed. And both of the authentication failure from the AS and the Key-auth verification failure make the STA abort this attempt and return to the initial step. With this mechanism, the STA will not accept more than one Msg2 from the AP. Instead, it only accepts the first valid Msg2, which can reduce the overhead for calculating new keys and installing the new key set.

AP has the similar operations with minor differences in Figure 4 in appendix. Since Msg1 is the initialization message, the AP can only broadcast before receiving Msg1. When Msg1 is received, the AP verifies the sequence number and appends ANonce to Msg1. After receiving Msg2', the AP temporarily accepts PMK and PTK from the AS used as Temporary PMK (TPMK) and Temporary PTK (TPTK). If the AP cannot receive Msg3 after sending Msg2, it regenerates the new ANonce as well as the key sets to ensure the freshness of new Msg2 and send it again. The reason for the AP to be able to calculate Msg2 again is that the AS has already sent rMSK and rIK to the AP. The first calculation of the key set done by the AS is to reduce the calculation burden to the AP and ensure efficiency. In practice, the AP can be in charge of hundreds or thousands of the STAs. Thus, it could save resources for the AP if the channels are stable enough so that only a small portion of the key sets will be regenerated by the AP. If any of the received Msg3 indicates the failed Key-auth field, the AP will return to the initial step waiting for a new Msg1. The operation will make sure that the AP will not return to any previous step after it receives more than one valid Msg3 so that the AP will not reinstall the key set to prevent the KRAs.

Since the SEF is designed to serve as a substitute to the FILS shared key authentication, it inherits the Perfect Forward Secrecy (PFS) of the FILS. It is allowed to indicate the SEF with or without PFS by broadcasting it in the AP's beacon or probe response of Msg1.

IV. Security Evaluation

To ensure the security functionality of the proposed SEF scheme, security evaluation is conducted. Burrows–Abadi–Needham (BAN) Logic is employed to ensure that there is no logical loophole could be taken use by attackers. Also the qualitative analysis of the ability of the SEF scheme against typical attacks is presented.

A. Proof of Logical Correctness

BAN Logic is applied to verify the logical correctness of the SEF scheme to ensure inexistence of logical loopholes. The idealized forms of messages include the encryption key and nonces. The first two messages are protected by rIK as specified in the ERP. The rIK is used rather than the rRK, because rIK is uniquely derived from rRK and the rRK is already shared, but never used to encrypt messages. Thus, the message form only includes rIK and rRK can be used as an assumptions. Besides, the last message is encrypted by newly generated PTK. Nonces included in the messages are used to calculate keys and provide the freshness of messages. Also, the AS has an indispensable role in this proof and serves as the bridge to connect the STA and the AP because rRK is shared between the STA and the AS, not between the STA and the AP. And the AP shares with the AS over a secure link. The postulates and symbol meanings were introduced in [9]. The STA and the AP are denoted by S and A, respectively.

- Idealized Messages

S→A: $\{SNonce, A \overset{rIK}{\leftrightarrow} S\}_{rIK}$

A→S: $\{ANonce, \{GTK, A \overset{PTK}{\leftrightarrow} S\}_{PTK}, A \overset{rIK}{\leftrightarrow} S\}_{rIK}$

S→A$\{Nonce, A \overset{PTK}{\leftrightarrow} S\}_{PTK}$

- Goals

These goals allow the AP to use rMSK to derive PMK and use it as a session key. They are denoted as follows:

$$A| \equiv S| \equiv AS \overset{rRK}{\leftrightarrow} S \qquad S| \equiv A| \equiv AS \overset{rRK}{\leftrightarrow} S$$

$$A| \equiv A \overset{PMK}{\leftrightarrow} S, \qquad A| \equiv S| \equiv A \overset{PMK}{\leftrightarrow} S$$

$$S| \equiv A \overset{PMK}{\leftrightarrow} S, \qquad S| \equiv A| \equiv A \overset{PMK}{\leftrightarrow} S$$

$$A| \equiv A \overset{PTK}{\leftrightarrow} S, \qquad A| \equiv S| \equiv A \overset{PTK}{\leftrightarrow} S$$

$$S| \equiv A \overset{PTK}{\leftrightarrow} S, \qquad S| \equiv A| \equiv A \overset{PTK}{\leftrightarrow} S$$

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The Group Temporal Key (GTK) has to be delivered to the STA and confirmed.

$$S| \equiv GTK, \qquad A| \equiv S| \equiv GTK$$

- Assumptions

They first have to believe in the previously shared key, rRK. Note that these two parties here are the STA and the AS.

$$AS| \equiv AS \overset{rRK}{\longleftrightarrow} S \qquad S| \equiv AS \overset{rRK}{\longleftrightarrow} S$$

Similarly, both parties, the STA and the AP, have to believe in the freshness of nonces distributed by each other.

$$A| \equiv \#\{SNonce\}$$

$$A| \equiv \#\{Nonce\}$$

$$S| \equiv \#\{ANonce\}$$

Because the GTK needs to be delivered, the STA has to believe that the AP has control over the GTK.

$$S| \equiv A| \Rightarrow GTK$$

- Verification

From Msg1, since $AS| \equiv AS \overset{rRK}{\longleftrightarrow} S$, and rIK is derived from rRK as mentioned before. We can derive

$$AS| \equiv AS \overset{rIK}{\longleftrightarrow} S$$

According to the message-meaning rule,

$$\frac{AS| \equiv AS \overset{rIK}{\longleftrightarrow} S, AS \lhd \{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}{AS| \equiv S| \sim \{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}$$

According to belief properties,

$$\frac{AS| \equiv \#\{SNonce\},}{AS| \equiv \#\{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}$$

According to the nonce-verification rule,

$$\frac{AS| \equiv \#\{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}, AS| \equiv S| \sim \{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}{AS| \equiv S| \equiv \{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}$$

According to belief properties,

$$\frac{AS| \equiv S| \equiv \{SNonce, AS \overset{rIK}{\longleftrightarrow} S\}}{AS| \equiv S| \equiv AS \overset{rIK}{\longleftrightarrow} S}$$

Theoretically, since rIK is uniquely derived from rRK, belief on rIK can be regarded as belief on rRK.

$$AS| \equiv S| \equiv AS \overset{rRK}{\longleftrightarrow} S$$

Since the AP trust what the AS sends to it, it believes what the AS believes.

$$A| \equiv S| \equiv AS \overset{rRK}{\longleftrightarrow} S$$

After receive rMSK from the AS, the AP generates its ANonce and use it to calculate PMK and PTK. Thus

$$A| \equiv A \overset{PTK}{\longleftrightarrow} S, \qquad A| \equiv A \overset{PMK}{\longleftrightarrow} S$$

From Msg2 sent by the AP, with the same postulates, the followings can be derived.

$$S| \equiv A| \equiv AS \overset{rRK}{\longleftrightarrow} S$$

$$S| \equiv A \overset{PTK}{\longleftrightarrow} S, \qquad S| \equiv A \overset{PMK}{\longleftrightarrow} S$$

By applying similar rules and postulates in the part encrypted by PTK, we can derive

$$S| \equiv A| \equiv GTK$$

$$S| \equiv A| \equiv A \overset{PTK}{\longleftrightarrow} S$$

According to jurisdiction rule,

$$\frac{S| \equiv A| \Rightarrow GTK, S| \equiv A| \equiv GTK}{S| \equiv GTK}$$

Similarly, PTK is also uniquely derived from PMK so that

$$S| \equiv A| \equiv A \overset{PMK}{\longleftrightarrow} S$$

With the same rules applied to Msg 3: S→A: $\{Nonce, A \overset{PTK}{\longleftrightarrow} S, \}_{PTK}$, it can be derived as follows:

$$A| \equiv S| \equiv A \overset{PMK}{\longleftrightarrow} S$$

$$A| \equiv S| \equiv A \overset{PTK}{\longleftrightarrow} S$$

$$A| \equiv S| \equiv GTK$$

After the verifications described as above, the logical correctness of the SEF scheme is verified. It not only confirms the possession of parent keys in the key hierarchy, but also proves that child session keys, such as PTK and GTK, are successfully derived. The confirmation of possession of parent keys allows the STA and the AS authenticate each other's identity. And the possession of child session keys allows the STA and the AP encrypt the data flow in the data transfer step.

B. Security Analysis

The SEF scheme is immune to several typical malicious attacks. The ability of the SEF scheme is explained as below.

The SEF scheme is designed to defend against the KRAs. There are two possible ways to launch a KRA. One is on the STA while another is on the AP. By the newly designed operations of the STA and the AP, it is able to ensure stable operations free from KRAs. For example, by the original FILS authentication, when the attacker tries to cheat the STA that the second Msg2 received is also a legitimate message, the STA cannot tell whether it is true or not and simply performs key installation again, which makes the same key installed twice. However, by the new operations of the SEF, the STA is able to distinguish the retransmitted key installation requests and strictly follow the secure procedure without installing the keys twice. Since the STA uses a Boolean value as the flag to indicate the installing state of the keys and the STA can only accepts the first valid Msg2 from the AP, it cannot only ensure that the STA would not be susceptible to reinstallation requests, but also decreases the overhead for calculating and installing new keys. The SEF scheme can also protect the AP in the similar way.

Inherited from the FILS, the protection on first message against the DoS attacks could be limited, because it depends on the time period the AS spends on the verification. Thus, the brutal DoS attacks may take some effects on the SEF. This could be a worth studying issue in the future. But the second and third messages can be both protected by the KEK scheme so that the attackers cannot flood the valid request without having the keys.

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To deal with replay attacks, all messages contain a sequence number. So long as the current sequence numbers are larger than the previous one, the both parties will accept them. Because these messages are all protected by MIC with AEAD so that the sequence numbers in the header cannot be changed by the attackers without being noticed. That is the reason for the KRAs to take the MitM position to control the retransmission and imitate replay attacks with the legitimate retransmitted message. Replaying the same message can be easily detected and discarded.

The MitM attacks are infeasible so long as the integrity and confidentiality are not compromised. The MitM attacks need to earn the trust from one party and disclosure of some information through the message exchange. Nevertheless, the MitM attackers have no way to acquire any of the root key or session key. Thus, the attacker would not be trusted or able to disclose information. It needs to be clarified that the KRAs need to take a middle position between two parties, which does not mean that that KRAs can be categorized as a type of MitM attacks, because the attackers can only faithfully relay the messages without altering messages or injecting any information into messages. Although the KRAs can intercept the replying message to lure the retransmitted message, they eventually relay both the original and retransmitted messages to the victim and will not change any single bit of them.

An impersonation attack is a type of malicious attacks by which an adversary can successfully pretend to be one of the legitimate parties in the system or in a communication protocol, thus receiving messages directed to the party it fakes [10]. The factor to make the impersonation attacks successful is that the attacker can obtain the trust from the network. By the SEF scheme, the STA and the AS have already shared a common rRK because they have completed the authentication before or the STA has already registered in the AS. Thus, any attackers wish to impersonate the STA will fail because they do not have a valid rRK. Since the AP sends the valid Key-auth field encrypted by valid PTK, which are derived from rMSK, a child key of rRK, the STA trusts the AP. While the fake AP is impossible to hold a valid rRK. Since the communication between the AP and the AS is assumed to be secure in the paper, the impersonation of the AS is out of scope.

Although obtaining key is a difficult and it is a resource-intensive process for any attackers, it is still possible for the attackers to compromise the STA and acquire the shared key rRK. However, since rRK has its life time, and all the keys derived from it must have their life times less than that of rRK.

Even if all keys generated from rRK are known to the attackers, this key set can only be used to decrypt the messages sent to the STA itself, which means that this common key set can only be used for this STA and the AP within the limited specified life time. The compromised attack could cause damage to the network, but it can be limited to the STA only.

V. Performance Evaluation

A. Cost Analysis

Besides the security enhancement achieved by the SEF scheme, it is also necessary to evaluate the performance of it to demonstrate that the SEF scheme can reduce the overheads in terms of the time delay and the energy consumption. In the section, we focus on the cost analysis of several typical cryptographically operations required by both of the FILS and SEF schemes, regardless of the overheads for information processing, transmitting, receiving, or installing key sets and so forth. Table 1 lists the operations comparison of two schemes. The security functions listed are necessary for the authentication and key generation. And in Table 2, the parameters for security functions are provided with the lengths of each of them. Since no system running 802.11ai protocol is deployed, the performance evaluation is carried out based on the parameters from the experiments in [11], where the experiments conducted by using a PC of Compaq iPAQ H3670, which uses an Intel SA-1110 StrongARM processor clocked at 206MHz. The final results can be calculated as in Table 3. The results show that the SEF scheme has significantly outperformed the FILS scheme in terms of 17.2% reduction on the time delay and 22.8% reduction on the energy consumption. Although the practical results may vary based on different platforms and environments, it clearly demonstrates that the SEF scheme is much more efficient than the FILS scheme. This achievement mainly comes from the reallocation of the calculation tasks by the SEF scheme.

Table 1 Operations Comparison

Message	FILS	SEF
S→A	SNonce generation, rIK derivation	SNonce generation, rIK derivation
A→S	ANonce generation, PMK derivation, PTK derivation, GTK derivation	ANonce generation, GTK derivation
S→A	rMSK derivation PMK derivation, PTK derivation, Key-auth hash,	PMK derivation, PTK derivation, Key-auth hash
A→S	Key-auth hash,	N.A

Table 2 Parameters for Security Functions

Parameters	Length (bit)
PMK	256
PTK	512
Nonce	256
GTK	128
MAC address	48
Sequence Number	128
rRK	512
rIK	512
rMSK	512

Table 3 Performance Comparison

Message	Time Delay(us)		Energy Consumption(mJ)	
	FILS	***SEF***	***FILS***	***SEF***
S→A	127.95	127.95	150.59	150.59
A→S	183.03	55.08	216.23	65.64
S→A	313.67	216.52	371.15	257.88
A→S	88.57	N.A	107.29	N.A
Total	603.22	499.55	705.26	544.11

B. Performance Evaluation by Simulation

We have also conducted a set of simulation experiments to further evaluate the performance of the both schemes under known and unknown attacks. The simulation experiments have been conducted by programming with Matlab 2014b carried out on the PC with its CPU of Intel(R) Xeon(R) CPU-E51650-v4 @3.6GHz. In total, there are four experiments carried out to

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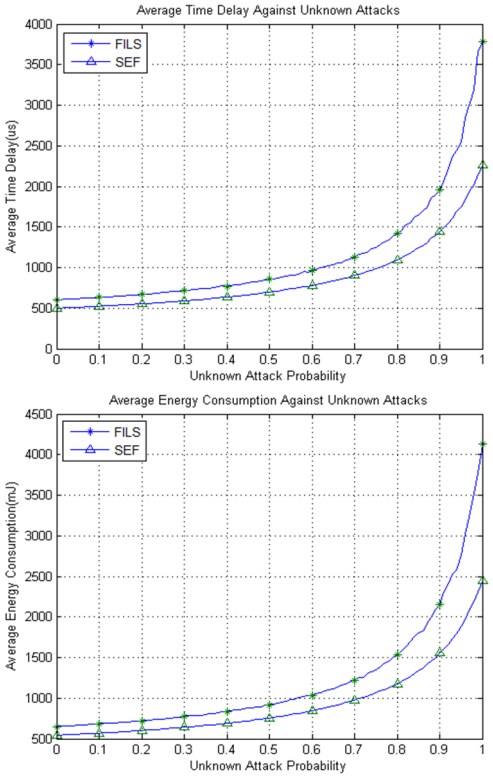


Figure 2 Average Delay and Energy Consumption

evaluate the performances of the FILS and the SEF schemes in terms of the time delay and energy consumption of the applied cryptographic algorithms, which are hash function SHA-256, symmetric encryption AES-128.

The simulation results reveal that the proposed SEF scheme outperforms the FILS scheme under the different types of malicious attacks because the SEF scheme has performed with less number of steps. The simulation results are illustrated in Figure 2. It is obvious that with the increasing probability of unknown attacks, the average time delay and energy consumption increase as well. More percentage of unknown attacks emerge during the simulation, which means that higher probability to interrupt the protocol execution within the period. The experiments end to get the curve with the similar indications. If the unknown attack probability is zero, there is no unknown attack launched so that the time delay and energy consumption are the same as those in the Table 3. If the probability of unknown attacks is 100%, the execution of the two schemes will always in the procedure of start and finish of the authentication process so that the consumption of both schemes reaches the highest point on average. It is clear that the differences between two schemes are not very distinct. However, it still demonstrates that the improvement on security will not degrade the performance of the networks.

VI. Conclusions

In this paper, as the pioneer research work to enhance the security of the FILS, we have designed and proposed the SEF scheme. First, the vulnerability of the FILS under the KRA has been explored followed by the motivation of our proposed solution. Then, we have presented our solution in details. Furthermore, we have conducted the security evaluation on the proposed SEF scheme including the proof of logical correctness followed by the qualitative security analysis. Finally, we have conducted the performance evaluations on the SEF scheme to clearly demonstrate that our proposed SEF scheme not only has strong security functionality with the ability against the KRAs, but also holds an outstanding performance with much more efficiency than that of the FILS scheme.

References

[1] IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems - Local and Metropolitan Area Networks--Specific requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 1: Fast Initial Link Setup," in IEEE Std 802.11ai-2016 (Amendment to IEEE Std 802.11-2016), Dec, 30 , 2016. pp.1-164.

[2] V. Jones and H. Sampath, "Emerging Technologies for WLAN," *IEEE Communications Magazine,* Vol. 53, No. 3, Mar, 2015. pp. 141-149.

[3] E. H. Ong, "Performance Analysis of Fast Initial Link Setup for IEEE 802.11 ai WLANs," *Proceedings of IEEE 23rd International Symposium on Personal Indoor and Mobile Radio Communications (PIMRC)*, Nov. 2012, pp. 1279-1284.

[4] X. Li, F. Bao, S. Li, and J. Ma, "FLAP: An efficient WLAN initial access authentication protocol," IEEE Transactions on Parallel and Distributed Systems, vol. 25, no. 2, pp. 488-497, 2014.

[5] S. Lee, B. H. Chung, " Denial of Service Attack against IEEE 802.11 WLAN Fast Initial Link Setup Technology," International Journal of Computer Science and Electronics Engineering (IJCSEE), vol. 3, no. 4, pp. 335-337, 2015.

[6] Z. Cao, B. He, Y. Shi, Q. Wu, and G. Zorn, "EAP Extensions for the EAP Re-authentication Protocol (ERP)," 2070-1721, 2012. *https://www.rfc-editor.org/info/rfc6696*

[7] M. Vanhoef and F. Piessens, "Key Reinstallation Attacks: Forcing Nonce Reuse in WPA2," *Proceedings of the ACM Conference on Computer and Communications Securit,* Vol. 30, Oct-Nov 2017, pp. 1313-1328.

[8] M. Vanhoef and F. Piessens, "Advanced Wi-Fi Attacks Using Commodity Hardware," *Proceedings of the 30th Annual Computer Security Applications Conference*, Dec, 2014, pp. 256-265.

[9] M. Burrows, M. Abadi, and R. M. Needham, "A logic of Authentication," *Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences*, Vol. 426, No. 1871, Dec, 1989, pp. 233-271.

[10] L. Tamilselvan and D. V. Sankaranarayanan, "Prevention of Impersonation Attack in Wireless Mobile ad hoc Networks," *International Journal of Computer Science and Network Securit,* Vol. 7, No. 3, Mar, 2007, pp. 118-123,

[11] N. R. Potlapally, S. Ravi, A. Raghunathan, and N. K. Jha, "A study of the energy consumption characteristics of cryptographic algorithms and security protocols," IEEE Transactions on mobile computing, vol. 5, no. 2, pp. 128-143, 2006.

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